

Variably irradiated tidally locked planets

DEEPAYAN BANIK ¹, VASIL PUSTOVOIT ¹, THADDEUS D. KOMACEK,² AND MARTA L. BRYAN ³

¹*Department of Physics, University of Toronto, Canada*

²*Department of Physics, Oxford University, United Kingdom*

³*Department of Astronomy, Pennsylvania State University, United States of America*

ABSTRACT

Temporally modulated stellar heating affects the climate of pseudo-synchronously rotating planets on eccentric orbits and those orbiting gravity-darkened stars. We study the atmospheric response of variably irradiated tidally locked exoplanets using a shallow water model forced toward a time-dependent radiative equilibrium. Linearizing the equations, we derive analytic expressions for the amplitude and phase lag of the atmospheric response in terms of dynamical parameters, including wave, radiative, and drag timescales, effectively reducing seven parameters to two. In the limit of steady forcing, our results recover known features of synchronously rotating planets, validating the model. We find close matches of the analytical results with linear and nonlinear simulations albeit with minor quantitative differences. At low drag/dissipation, when the wave timescale matches the radiative timescale, the hotspot location shows periodic oscillations corresponding to the variation of instellation moving between the substellar and antistellar points, having imprints on the phase curve of such planets.

Keywords: Galaxies (573) — Cosmology (343) — High Energy astrophysics (739) — Interstellar medium (847) — Stellar astronomy (1583) — Solar physics (1476)

1. INTRODUCTION

Many close-in exoplanets are expected to be *tidally locked* to their stars, so that one hemisphere receives continuous stellar irradiation while the opposite hemisphere remains in perpetual darkness. In such cases a permanent dayside and nightside develop, leading to extreme longitudinal heating contrasts (R. Pierrehumbert & M. Hammond 2019). This intense day–night thermal forcing drives a characteristic circulation: broad equatorial superrotation and planetary-scale standing waves arise to transport heat. Analytical and numerical models show that standing equatorial waves forced by the day–night heating can pump eastward momentum into the equator, naturally producing a wide superrotating jet and an eastward-shifted thermal hotspot (A. P. Showman & L. M. Polvani 2011; D. Perez-Becker & A. P. Showman 2013a). For example, Perez-Becker & Showman (2013) used a two-layer shallow-water model to demonstrate that strong stellar forcing (short radiative time) leads to nearly direct day–night flow with order-unity temperature contrasts, whereas weaker forc-

ing yields banded zonal flows with smaller contrasts (D. Perez-Becker & A. P. Showman 2013a). These results are borne out by 3D circulation models and reviewed in detail (R. Pierrehumbert & M. Hammond 2019).

However, this picture assumes constant stellar forcing. In many systems the irradiation varies periodically. Examples include:

- **Eccentric orbits / pseudo-synchronous spin:** HD 80606b ($e \approx 0.93$) receives short, intense bursts of heating near each periastron (J. Langton & G. Laughlin 2008).
- **Gravity-darkened stars:** KELT-9 is a very rapidly rotating star whose poles are much hotter than its equator (von Zeipel effect), so KELT-9b sees a latitude-dependent flux pattern that rotates with the star (K. D. Jones et al. 2022).
- **Pulsating/variable stars:** WASP-33b orbits a δ -Scuti pulsator, causing its incident flux to oscillate on timescales comparable to the orbital period (C. von Essen et al. 2020).

In each of these cases the pattern of heating on the planet changes over time, even if the planet’s spin re-

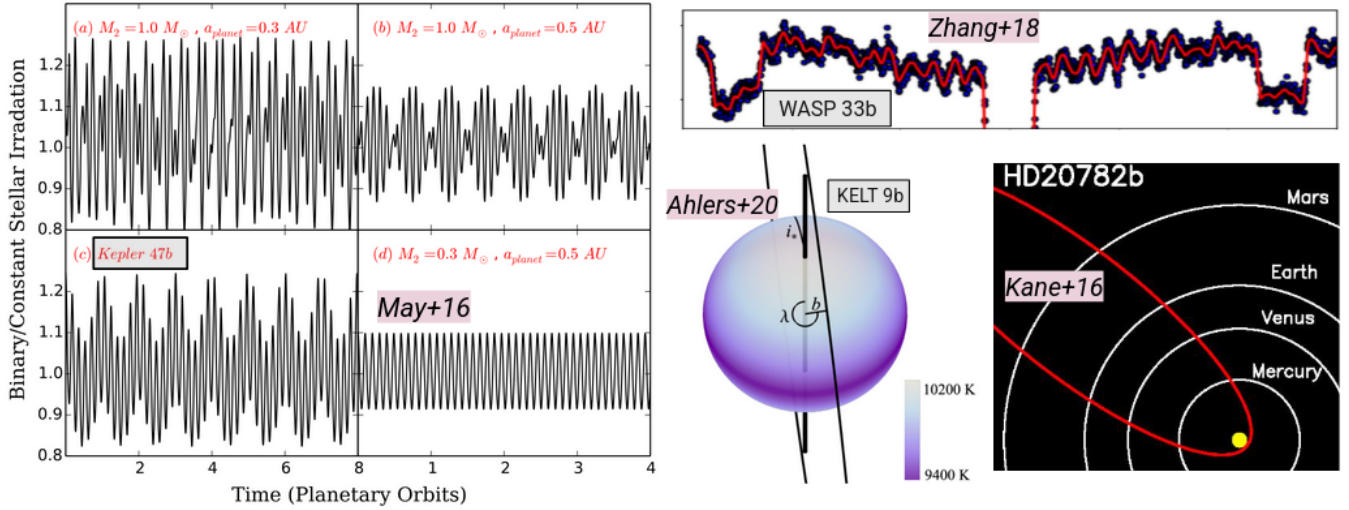


Figure 1. Different kinds of variably irradiated systems, including circumbinary planets (Kepler-47b), tidally-locked planets around variable stars like the Delta Scuti variables (WASP-33b), planets in polar orbits around gravity darkened stars (KELT-9b), and eccentric/pseudo-synchronous planets (HD20782b).

mains synchronized on average. This time-dependent forcing can significantly modify the atmospheric winds and temperature distribution compared to the steady case, and thus the observed phase curve.

Several studies have begun to explore time-varying irradiation. GCM simulations have found that eccentric hot Jupiters still develop strong equatorial jets, but the timing of peak infrared emission can shift relative to periastron (showing “ringing” in phase curves) depending on orbital geometry (T. Kataria et al. 2013). Compressible shallow-water simulations showed that HD 80606b’s atmosphere is dominated by the periastron heating pulse, driving high-speed turbulent flows and long-lived vortices (J. Langton & G. Laughlin 2008). Recently, a shallow-water model was used to demonstrate a resonant “beating” effect on asynchronous planets: if the installation variation period nearly matches the planet’s day, the hotspot can lock to a fixed longitude (D. Banik 2025). However, a simple analytic theory for the oscillatory response is still lacking.

In this work we develop such a model. We construct a linear two-dimensional shallow-water system on an f -plane (representing mid-latitude dynamics) forced by a sinusoidally varying radiative-equilibrium height field. The steady-state oscillatory solution can be derived in closed form, yielding explicit expressions for the amplitude and phase lag of the height (temperature) and wind perturbations as functions of the forcing frequency and physical parameters (rotation rate, radiative timescale, drag, etc.). We validate the linear theory against fully nonlinear global simulations (using the Dedalus spectral solver; K. J. Burns et al. 2020a), finding good agreement. In the limit of zero forcing frequency our solution

reduces to the standard tidally locked solution of Perez-Becker & Showman (2013) (D. Perez-Becker & A. P. Showman 2013a). Finally, we apply the model to two illustrative cases—a pseudo-synchronous eccentric planet (HD 80606b) and a planet around a gravity-darkened star (KELT-9b)—showing how time-variable forcing alters the heat redistribution relative to the constant-irradiation baseline.

2. THEORIZING WITH LLMS

3. LINEARIZED SHALLOW WATER THEORY

We utilize the same spherical (ϕ, λ) two-layer shallow water model as in D. Banik (2025) to simulate the atmosphere of variably irradiated tidally-locked planets. For details the reader is referred to that paper. The shallow water model offers the benefit of efficient parameter space survey in addition to capturing leading order circulation physics, and has been extensively used in literature before (D. Perez-Becker & A. P. Showman 2013b; K. Ohno & X. Zhang 2019; J. Penn & G. K. Vallis 2017). The model features an active upper layer $h(\lambda, \phi, t)$ over an inert layer of infinite depth. The governing equations for momentum and mass balance are,

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} + f \mathbf{k} \times \mathbf{u} + g \nabla h = \mathbf{R} - \frac{\mathbf{u}}{\tau_{\text{drag}}} \quad (1)$$

$$\frac{\partial h}{\partial t} + \nabla \cdot (h \mathbf{u}) = \frac{h_{\text{eq}} - h}{\tau_{\text{rad}}} \equiv Q \quad (2)$$

$$\mathbf{R} = -\frac{\mathbf{u}}{h} \cdot \frac{Q + |Q|}{2} \quad (3)$$

where g is the acceleration due to gravity on the planet, $\mathbf{u}$ is horizontal velocity, $f = 2\Omega \sin \phi$ is the Cori-

olis parameter, τ_{drag} is the Rayleigh drag timescale, and τ_{rad} is the Newtonian cooling timescale. The term $\mathbf{R}$ ensures conserved mass/momentum exchanges between the two layers, emergence of equatorial superrotation, and ensures independence from initial conditions (A. P. Showman & L. M. Polvani 2010; A. P. Showman & L. Polvani 2011; B. Liu & A. P. Showman 2013). Fluid height acts as a proxy for temperature.

3.1. Steady-state Scaling Analysis

We carry out scaling analysis similar to D. Banik et al. (2025). For that, we linearize the system with the following assumptions:

- The basic state are zero background flow for velocity $\mathbf{u}_0 = 0$ and the nightside height H for height.
- Small perturbations are added on top of the base state i.e. $\mathbf{u} = 0 + \mathbf{u}'$, $h = H + h'$, with $|\mathbf{u}'| \ll \sqrt{gH} = c$, or the gravity wave speed of the shallow atmosphere, and $|h'| \ll H$.
- We decompose $h_{\text{eq}} = H + h'_{\text{eq}}$, not for the purpose of linearizing but to express the equation in terms of the variable h'_{eq} , the deviation from background height.
- We also neglect the nonlinear vertical exchange term $\mathbf{R}$ to simplify the system.
- Finally, we combine the Coriolis and drag terms to form the effective drag approximation ($\mathbf{k} \times \mathbf{u}' \approx \sin \phi \mathbf{u}'$, $\sin \phi \approx \sin \phi_0$, $\tau_{\text{eff}}^{-1} = f \sin \phi_0 + \tau_{\text{drag}}^{-1}$). Thus, the combined term is $\tau_{\text{eff}}^{-1} \mathbf{u}'$. We acknowledge that rotational and drag effects can be dramatically different especially considering latitudinal variations. However, at first order both terms contribute similarly to heat transport (D. Perez-Becker & A. P. Showman 2013b; D. Banik 2025).

We first derive the scales for the unknown quantities height and velocity from the linearized treatment of the steady-state by dropping the time derivatives. Then we use these to derive conditions for nonlinearity through order of magnitude comparisons. We define the following characteristic scales: horizontal length $L \sim a$ (planetary radius), velocity U , height perturbation Δh , forcing Δh_{eq} , and nightside thickness H . The linearized equations of motion produce the following momentum and mass balances,

$$0 = -g\nabla h' - \frac{\mathbf{u}'}{\tau_{\text{eff}}} \implies 0 \sim -g\frac{\Delta h}{a} - \frac{U}{\tau_{\text{eff}}} \quad (4)$$

$$0 = -H\nabla \cdot \mathbf{u}' + \frac{h'_{\text{eq}} - h'}{\tau_{\text{rad}}} \implies 0 \sim -H\frac{U}{a} + \frac{\Delta h_{\text{eq}} - \Delta h}{\tau_{\text{rad}}} \quad (5)$$

From the momentum balance (4), we find the velocity scale as,

$$U \sim \frac{g\Delta h\tau_{\text{eff}}}{a} \quad (6)$$

We use the gravity wave speed $c = \sqrt{gH}$ to define the wave timescale as $\tau_{\text{wave}} = a/c$. Substituting the above into the continuity balance (5) we get

$$\frac{\Delta h}{\Delta h_{\text{eq}}} \sim \left(1 + \frac{\tau_{\text{rad}}\tau_{\text{eff}}}{\tau_{\text{wave}}^2}\right)^{-1} \text{ or } \Delta h = \frac{\Delta h_{\text{eq}}}{1 + 1/K} \quad (7)$$

where $K = \tau_{\text{wave}}^2/(\tau_{\text{rad}}\tau_{\text{eff}})$. Thus, we see that the global contrast Δh is dependent only on two parameters, the forcing Δh_{eq} and K , the dimensionless ratio of wave and damping (radiative and momentum) timescales. This is commensurate with the scaling law for hot Jupiters in D. Perez-Becker & A. P. Showman (2013b), except it is a more condensed version combining the effects of rotation and drag in a simplified form. The characteristic velocity U is of the form,

$$U \sim \frac{a\tau_{\text{eff}}}{H\tau_{\text{wave}}^2} \cdot \frac{\Delta h_{\text{eq}}}{1 + 1/K} \quad (8)$$

This is the velocity scaling corresponding purely to thermal forcing, and has a similar structure to that of momentum forcing derived in T. D. Komacek (2025).

3.1.1. Conditions for linearity

The linear approximation holds only when the nonlinear terms in the momentum and continuity equations are negligible compared to the corresponding linear terms.

- Height Linearity Condition: The thickness advection $\nabla \cdot (h\mathbf{u}')$ which scales as $\Delta hU/a$ must be small compared to the mean mass flux $H\nabla \cdot \mathbf{u}' \sim HU/a$. This is satisfied when

$$\frac{\Delta h}{H} = \frac{\Delta h_{\text{eq}}/H}{1 + 1/K} \ll 1 \quad (9)$$

- Momentum Linearity Condition: From the above scalings, the advection term $(\mathbf{u}' \cdot \nabla)\mathbf{u}'$ scales as U^2/a and must be small compared to that of drag U/τ_{eff} . Substituting U from (8) we get,

$$\frac{U\tau_{\text{eff}}}{a} = \frac{\tau_{\text{eff}}^2}{\tau_{\text{wave}}^2} \cdot \frac{\Delta h_{\text{eq}}/H}{1 + 1/K} \ll 1 \quad (10)$$

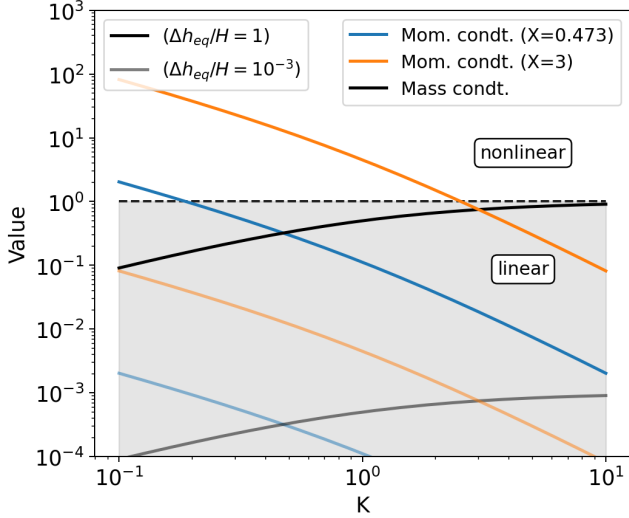


Figure 2. Magnitude of linearity constants in inequalities (10) and, (9) with K for different system parameters assumed in this work. Note that $X = \tau_{\text{wave}}/\tau_{\text{rad}}$.

In a different, but familiar geophysical fluid dynamics context, this corresponds to the “frictional” Rossby number (rotation plus drag) being smaller than 1, such that inertia/ material acceleration terms cannot dominate the dynamics.

D. Perez-Becker & A. P. Showman (2013b), henceforth also referred to as PBS13, set the ratio of forcing to the nightside height ($\Delta h_{\text{eq}}/H$) to 0.001 and 1, to cover linear and nonlinear regimes of operation. However, that is not sufficient to ensure either linear or nonlinear behavior. Both (10) and, (9) must be satisfied, out of which the momentum condition, unlike the mass condition, depends on $\tau_{\text{eff}}^2/\tau_{\text{wave}}^2$ in addition to the forcing Δh_{eq} and K . These conditions have not been previously derived for a forced shallow water model of a tidally-locked planetary atmosphere. Figure 2 plots the LHS of the linearity conditions as a function of K for two different values of $\Delta h_{\text{eq}}/H$, and two cases of planetary parameters, one corresponding the typical hot Jupiter assumed in D. Perez-Becker & A. P. Showman (2013b) for which $X = 3$, and the other assumed arbitrarily for this work. Here, $X = \tau_{\text{wave}}/\tau_{\text{rad}}$. Comparing figures 3 and 4 in PBS13, we find that the difference between linear and nonlinear flows is more pronounced at low drag, which here corresponds to low K (high τ_{drag}) values, making their results agree with our scalings.

Figure 3 shows how the nondimensional forcing amplitude $\Delta h_{\text{eq}}/H$ varies as a function of the parameter K for a suite of numerical experiments spanning Earth-like and hot-Jupiter-like parameter choices performed later in the paper. Data points correspond to simulation re-

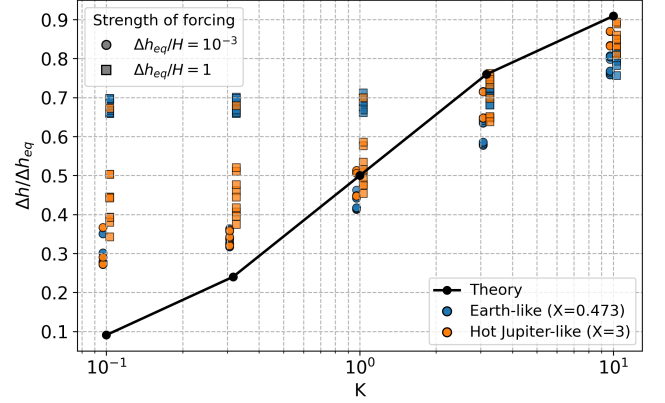


Figure 3. Variation of $\Delta h_{\text{eq}}/H$ with K for different Earth-like and Hot Jupiter planetary system parameters from all simulations performed in the following section compared with theoretical predictions. The linear (circles) and nonlinear (squares) regime simulations are horizontally offset by small quantity for the ease of visualization. Linear regime simulations match the theory very well, while the nonlinear regime simulations show quantitative differences, as expected.

sults (circles for the linear regime, squares for nonlinear runs) while the solid lines show the theoretical prediction from the linearized shallow-water balance. The plot demonstrates good agreement between theory and simulations in the linear regime, and systematic departures in the nonlinear cases especially at low K (weak drag or rotation), highlighting the role of nonlinearity in enhancing the height contrast. (What does PBS13 other literature say?)

3.2. Linearized first-order dynamics

When both the above conditions are satisfied, we may drop the corresponding nonlinear terms. In this section we consider the linearized limit in an attempt to find analytical solutions to relevant quantities like temperature contrast and average wind speed. Since we are investigating the effect of changing irradiation on the atmospheric height we must retain the unsteady term in equation (2). We begin with the assumption of a balanced flow originating from equation (1) where the pressure gradient is balanced by effective drag term. Ignoring nonlinear advection and unsteady terms gives us an equation similar to (4).

$$\frac{1}{\tau_{\text{eff}}} \mathbf{u}' + g \nabla h' = \mathbf{0} \implies \mathbf{u}' = -g \tau_{\text{eff}} \nabla h'. \quad (11)$$

Keeping $\partial_t h'$ in the mass balance and linearizing yields

$$\frac{\partial h'}{\partial t} + H \nabla \cdot \mathbf{u}' = \frac{h'_{\text{eq}} - h'}{\tau_{\text{rad}}}. \quad (12)$$

From this linearized height equation, we may compare the scales of the unsteady and the height advection (mass flux) terms to find the following timescale for the flow in general,

$$\frac{\Delta h}{\tau_o} \sim \frac{HU}{a} \implies \tau_o \sim \tau_{adv} \cdot \frac{\Delta h}{H} = \frac{\tau_{wave}^2}{\tau_{eff}} \quad (13)$$

where the advection timescale ($\tau_{adv} = a/U$) has been modified by the linearity constant in (9).

Taking divergence on both sides of equation (11) we get $\nabla \cdot \mathbf{u}' = -g \tau_{eff} \nabla^2 h'$ which can then be substituted in equation (12) to obtain the following PDE for height,

$$\frac{\partial h'}{\partial t} - gH \tau_{eff} \nabla^2 h' + \frac{h'}{\tau_{rad}} = \frac{h'_{eq}(\lambda, \phi, t)}{\tau_{rad}}. \quad (14)$$

This is a generalized first order equation in time representing the dynamics of planetary response to any form of spatially and temporally varying radiation. Next we shall investigate the behavior of the system.

Let us assume invariance in the meridional direction (ϕ) and reduce the spatial dimension of the problem to one. For simplicity, we shall also move to arc-length coordinates and replace λ by x . Additionally, we assume the forcing profile to be a simple cosine in x as follows $h'_{eq} = \Delta h_{eq} f(t) \cos(kx)$ (Add figure showing lat, lon, cosine, projection etc.), where $f(t)$ is some function of time.

We seek solutions of the form $h'(x) = \Delta h(t) \cos(kx)$, where $\Delta h(t)$ is the time-varying amplitude to be determined. This solution exists under the assumption of pure substellar to antistellar point flow. This results in the following linear damped forced ODE for the planetary heat/height contrast,

$$d_t \Delta h + \left(\frac{1}{\tau_{rad}} + gHk^2 \tau_{eff} \right) \Delta h = \frac{\Delta h_{eq} f(t)}{\tau_{rad}} \quad (15)$$

Having solved for Δh , we may substitute h' in equation (11) to uniquely determine the magnitude of the velocity field. Note that the velocity field varies as a sine from the substellar point due to the gradient operator on h' , implying that the highest and lowest temperature (max/min h') points or the hotspot and the coldspot are regions of stagnation and the winds achieves maximum value near the day-night divide i.e. the terminator.

If we take $f(t)$ to be unity and look for steady-state solutions, we recover the relation between planetary response and forcing as obtained from scaling relations in equation (8) which is as expected.

3.2.1. Variable instellation

In this section, we shall focus on systems with variable instellation. Even though the real nature of variability

can be significantly complicated as shown in figure 1, we start with simpler forms. We assume the forcing to be a simple sine superimposed on a background value and set $f(t) = (1 + F \sin \omega t)$, where ω is the instellation variation frequency. Beyond the transient, we obtain solutions of the form, $\Delta h(t) = \Delta h_0 + \Delta h_{amp} \cos(\omega t - \phi)$ to get,

$$\frac{\Delta h_0}{\Delta h_{eq}} = \left(1 + \frac{\tau_{rad} \tau_{eff}}{\tau_{wave}^2} \right)^{-1} = \frac{1}{1 + 1/K}, \quad (16)$$

$$\begin{aligned} \frac{\Delta h_{amp}}{\Delta h_{eq}} &= F \left[\left(1 + \frac{\tau_{rad} \tau_{eff}}{\tau_{wave}^2} \right)^2 + (\omega \tau_{rad})^2 \right]^{-1/2} \\ &= F \left[(1 + 1/K)^2 + W^2 \right]^{-1/2}, \text{ and} \end{aligned} \quad (17)$$

$$\phi = \tan^{-1} \left(\frac{\omega \tau_{rad}}{1 + \frac{\tau_{rad} \tau_{eff}}{\tau_{wave}^2}} \right) = \tan^{-1} \left(\frac{W}{1 + 1/K} \right). \quad (18)$$

where $W = \omega \tau_{rad}$ is the nondimensional instellation variation frequency. We also define some other nondimensional parameters $X = \tau_{wave}/\tau_{rad}$, $Y = \tau_{wave}/\tau_{drag}$, and $Z = a/L_d$ representing radiative, drag and rotational strengths as relevant to the global flow. Note here that $K = X(Y + Z)$. We see that the final expressions crucially depend on two parameters K and W only. Thus, the dimensionality of the parameter space is reduced from seven dimensional parameters ($f, g, a, H, \tau_{rad}, \tau_{drag}$, and ω) to two just nondimensional parameters. Here, K represents the planet's inherent capacity to prevent or allow heat redistribution, while W is the nondimensional frequency of instellation variation.

The first three panels in figure 4 show the magnitude, amplitude and phase of the planetary response to instellation variation in the $K - W$ space. The constant part of the planetary contrast is the same as obtained before and is unaffected by a change in ω . A large value of K corresponds to a planet that is strongly dragged, fast rotating, strongly radiated or any combination of them, all of which helps prevent heat redistribution. For the rest of the paper, we shall refer to these planets as strongly dragged only, for convenience. On the contrary, a planet with fast wave physics has a low K , and has better heat redistribution leading to an erasure of contrasts. The constant part of the contrast normalized by the forcing amplitude is thus seen to increase with K peaking at 1.

The amplitude of variation in planetary temperature behaves in a similar manner, except it is also affected by the frequency of instellation variation. When W or ω is large, the instellation fluctuates fast and the planet does not get enough time to adjust to the radiation leading

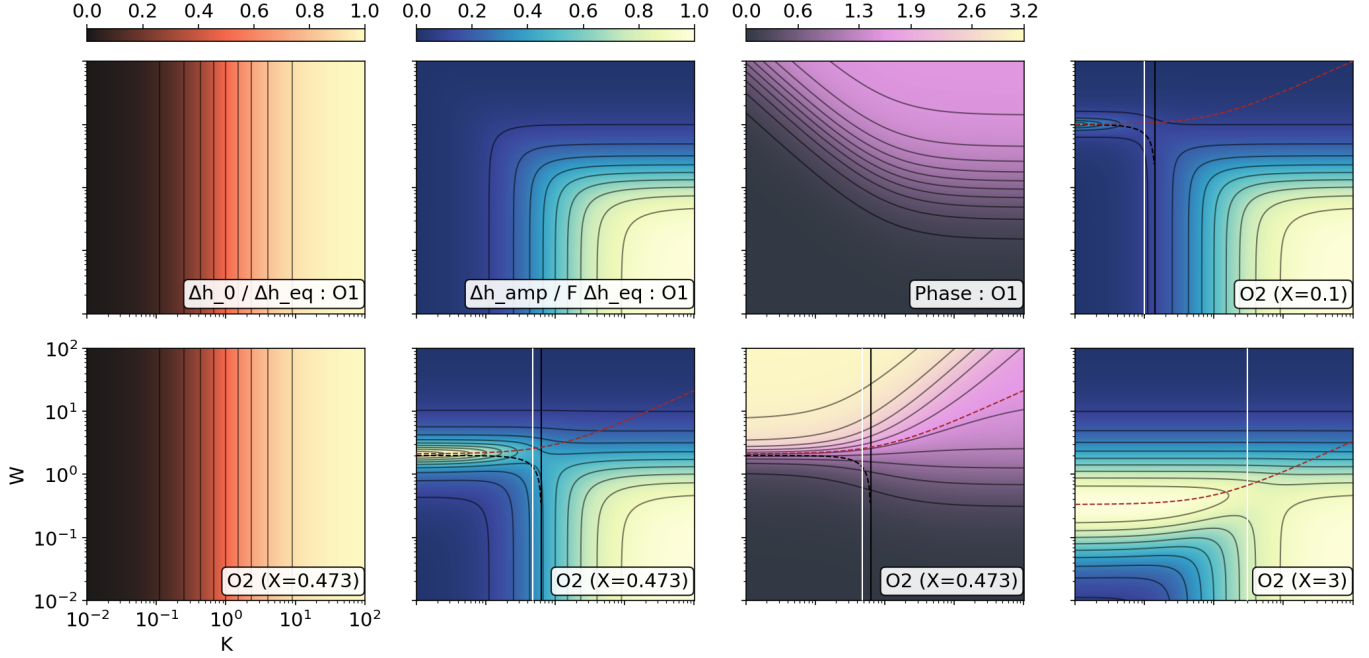


Figure 4. First and second order response of tidally-locked planets to variable heating, showing forcing normalized magnitude ($\Delta h_o/\Delta h_{eq}$), amplitude ($\Delta h_{amp}/F\Delta h_o$), and phase lag of the response using different colorschemes. The first order (O1) is dependent only on two nondimensional parameters $W = \omega\tau_{rad}$ and K , while the second order (O2) has a third parameter $X = \tau_{wave}/\tau_{rad}$. The magnitude responses for both are the same and commensurate with [D. Perez-Becker & A. P. Showman \(2013a\)](#). The brown and black dashed lines correspond to natural and resonant frequencies of the system. The solid black line corresponds to the critical damping K , to the left of which oscillatory solutions exist, and the system is overdamped to the right. The solid white line represents separation between the wave dominated inertial regime to the left of it and the drag/rotation dominated regime to the right, where the solutions match the first-order theory.

to a muted amplitude irrespective of the value of K , while low W allows the system to respond appropriately, leading to higher amplitudes.

Finally, the time phase lag of the planetary response is the highest ($= \pi/2$, the theoretical maximum) for strongly dragged systems under rapid instellation variation. Note that this phase is not the hotspot offset from the substellar point but the time difference between the peak of instellation and planetary maxima. More discussion on the hotspot offset and associated variability is included in a later section.

3.3. Linearized second-order dynamics

While the above first-order treatment with balanced flows is the most minimal model to capture time-variability, we recognize that the unsteady term in the momentum equation (1) need not be negligible. In fact, the condition for the unsteady term to be significant may be obtained by setting the scale for the term to be much greater than the effective drag term as below,

$$\frac{U}{\tau_o} \gg \frac{U}{\tau_{eff}} \quad \text{or} \quad \tau_o \ll \tau_{eff} \quad (19)$$

Substituting the timescale from equation (13), we get the condition to include the unsteady term as,

$$\tau_{wave} \ll \tau_{eff} \quad (20)$$

This implies that the drag or rotation of the system must be weak enough such that waves can dominate. In the opposite limit, gravity wave perturbations generated by heating are quickly damped preventing their effect from reverberating across the planet. When waves dominate new behavior emerges through constructive or destructive interference leading to resonances, as we shall see further. Assuming a cosine spatial structure along latitudes as before, we now simplify the mass and momentum balances keeping both unsteady terms to arrive at a second order ODE for the height as follows,

$$\begin{aligned} \ddot{\Delta h} + \left(\frac{1}{\tau_{rad}} + \frac{1}{\tau_{eff}} \right) \dot{\Delta h} + \left(\frac{1}{\tau_{wave}^2} + \frac{1}{\tau_{rad}\tau_{eff}} \right) \Delta h \\ = \Delta h_{eq} \left[\frac{1}{\tau_{rad}} \dot{f}(t) + \frac{1}{\tau_{rad}\tau_{eff}} f(t) \right]. \end{aligned} \quad (21)$$

We compare this equation with $\ddot{\Delta h} + \gamma \dot{\Delta h} + \Omega_o^2 \Delta h = F(t)$, which is the equation for a forced damped harmonic oscillator. This equation admits wave like solutions in time even without a sinusoidal forcing and has a

natural frequency $\Omega_o = \sqrt{\tau_{\text{wave}}^{-2} + \tau_{\text{rad}}^{-1}\tau_{\text{eff}}^{-1}}$, and damping $\gamma = \tau_{\text{rad}}^{-1} + \tau_{\text{eff}}^{-1}$. The constant part of the solution remains the same as before, but the amplitude and phase are modified as follows,

$$\frac{1}{F} \cdot \frac{\Delta h_{\text{amp}}}{\Delta h_{\text{eq}}} = \frac{\frac{1}{\tau_{\text{rad}}} \sqrt{\frac{1}{\tau_{\text{eff}}^2} + \omega^2}}{\sqrt{(\Omega_o^2 - \omega^2)^2 + (\gamma\omega)^2}} = \frac{\sqrt{K^2 + W^2 X^4}}{\sqrt{(1 + K - W^2 X^2)^2 + W^2(X^2 + K)^2}} \quad (22)$$

$$\tan \phi = \frac{\gamma\omega}{\Omega_o^2 - \omega^2} = \frac{W(X^2 + K)}{1 + K - W^2 X^2} \quad (23)$$

Note that in this case, an additional independent dimensionless parameter (X) is required to define the space of the problem contrary to the two (K and W) in the first-order theory. The resonant frequency of the system is

$$\Omega_{\text{res}} = \sqrt{\frac{1}{\tau_{\text{wave}}^2} + \frac{1}{\tau_{\text{rad}}\tau_{\text{eff}}} - \frac{1}{2} \left(\frac{1}{\tau_{\text{rad}}} + \frac{1}{\tau_{\text{eff}}} \right)^2}$$

$$W_{\text{res}} = \Omega_{\text{res}}\tau_{\text{rad}} = \sqrt{\frac{1}{X^2} + \frac{K}{X^2} - \frac{1}{2} \left(1 + \frac{K}{X^2} \right)^2} \quad (24)$$

The last first-row and the second-row panels of figure 4 show the magnitude, amplitude and, phase of planetary response for the second-order theory for different values of X . The magnitude of the response remains the same as captured by previous lower order theory and scaling relations, while the amplitude and phase show significant differences. The first thing to note is the region in $K - W$ space where the inertial physics (second-order theory) dominates. This is demarcated by the solid white line $K = X$, which is the condition derived from equation (20) by equating the wave and effective drag timescales. Comparing with the plots above, we notice that the solutions differ to the left of the white line confirming the inertial regime. This line closely follows the black solid line that separates regions having oscillatory versus overdamped solutions. Note that for $X = 3$, the entire parameter space is overdamped. Second, the inertial regime is seen to feature a local maxima in frequency space; this corresponds to the resonance of the forced damped oscillator. This frequency, and the spread of the maxima is dependent on the value of X as shown in the three blue plots in figure 4. The black and the brown dashed lines represent the resonant and natural frequencies of the system. Finally, the phase lag of the system can now take up values larger than $\pi/2$ as the theoretical limit reaches π symbolizing a complete

out-of-phase response. This is typical to the behavior of second order systems with a restoring force.

Thus, the response of tidally locked planets to time-variable irradiation can be described at two levels depending on whether inertia is dynamically active. Retaining local acceleration in the momentum equation leads to a formally second-order system in which the planetary heat contrast Δh behaves as a damped harmonic oscillator. In this regime, gravity-wave adjustment provides a restoring force while radiative relaxation and drag primarily provide damping.

In nonlinear simulations and likely in real atmospheres, gravity-wave adjustment is rapid and strongly damped by shocks, wave breaking, turbulence, and numerical diffusion. The system rapidly collapses onto a balanced slow manifold in which momentum is diagnostically determined by pressure gradients and drag. The resulting dynamics are thus first order in time. We shall test this hypothesis using nonlinear simulations in the next section.

3.4. Where do planets fall in $K - W$ space?

4. NUMERICAL SIMULATIONS

To evaluate the efficacy of our linearized theory, we compare the analytical results presented before to linear and nonlinear hydrodynamic simulations using the full shallow water model.

4.1. Framework

Equations (1) and (2) are solved numerically using Dedalus3 (K. J. Burns et al. 2020b), a flexible pseudo-spectral code for symbolic equation entry. We discretize the 2-D domain on a sphere basis and evolve the system using an SBDF2 timestepper. The simulation resolution is 32×16 (longitude $\times$ latitude), which, while low, sufficiently captures the core physical phenomena. Our code was verified by reproducing results from existing literature (D. Perez-Becker & A. P. Showman 2013b; J. Penn & G. K. Vallis 2017; K. Ohno & X. Zhang 2019) and is the same used in D. Banik (2025).

The tidally locked height field is relaxed toward a time-variable equilibrium profile h_{eq} to implement instellation variability.

$$h_{\text{eq}}(\lambda, \phi, t) = H + \Delta h_{\text{eq}} \cdot \frac{\mathcal{R} + |\mathcal{R}|}{2} (1 + F \cos \omega t)$$

where $\mathcal{R} = \cos \lambda \cos \phi$. (25)

Here, H is the nightside thickness, Δh_{eq} is the forcing day-night contrast, and F and ω are the instellation fluctuation amplitude and frequency. The factor $(\mathcal{R} + |\mathcal{R}|)/2$ isolates hemispheric irradiation (A. R. Dobrovolskis 2009).

Table 1. Physical parameters and timescales corresponding to the simulations presented. The table shows a total of 100 simulations, each of which are conducted at two different values of $\Delta h_{\text{eq}} = 10^{-3}H$ and H , representing the linear and nonlinear regimes described above, resulting in a total of 200 simulations.

Parameter	Symbol	Earth-like	Hot-Jupiter-like
Surface gravity	g	$9.81 \text{ m} \cdot \text{s}^{-2}$	$10 \text{ m} \cdot \text{s}^{-2}$
Scale height	H	100 m	$4 \times 10^5 \text{ m}$
Radiative timescale	τ_{rad}	5 days	0.1 days
Instellation variation	τ_{instel}	3.14 – 314 days ($\times 5$)	0.06 – 6.28 days ($\times 5$)
Wave timescale (derived)	τ_{wave}	2.365 days	0.4745 days
Scenario	Rotation Period (P_{rot})	Drag Timescale (τ_{drag})	No. of sims
A: Earth-like			
- Batch 1	140 days	0.11 – 22.373 days ($\times 5$)	25
- Batch 2	0.7 – 140 days	22.373 days ($\times 5$)	25
B: Hot-Jupiter			
- Batch 3	282 days	0.22 – 45 days ($\times 5$)	25
- Batch 4	1.42 – 282 days	45 days ($\times 5$)	25

4.2. Diagnostics

Each run produces timeseries of global height and velocity fields. These are post processed to obtain the quantities that can be compared with the theory above. The global height field is reduced to three quantities, the maximum height, the mean height and, the hemispheric averaged height. The last one is an average of the height field over a longitudinal extent of π (one face of the planet) for each individual longitude, and is a more accurate metric when comparing with observations that average the light coming from the exposed planetary hemisphere. The magnitude of planetary response is calculated by taking the average of the peak and trough of the maximum height timeseries after the simulation is out of its transient evolution. The amplitude is modulus half the difference between the peak and trough of the same. Finally, the phase is obtained by calculating the temporal offset between the peaks in instellation and mean height and converting it to radians. The reason for choosing the mean height here is that it has a smoother sinusoidal structure than maximum height, especially for inertia dominated cases where interacting waves interfere to create complicated responses.

We also obtain the longitudinal location of maximum height and the maximum hemispheric averaged height to track the motion of the real and observed hotspots. The details of this is included in the next section.

4.3. Exploration parameters

We simulate 25 planets that span the entire nondimensional $K - W$ parameter space sampling 5 K values for each of 5 W values. We use two different magni-

tudes for the forcing $\Delta h_{\text{eq}} = 10^{-3}$ and 1, respectively. F , the fractional amplitude is kept at 0.5. These parameter values cover the linear and nonlinear regime completely. We run these in two batches with $X = 0.528$ and 3, defining different ratios of radiative and wave adjustment strengths, as in figure 2. The first corresponds to smaller Earth-like planets, while the later uses the parameter set in D. Perez-Becker & A. P. Showman (2013b) applicable to hot Jupiters.

Here, we recall that in the linearized theory we have assumed the Coriolis and drag terms to be combined into the effective drag term for simplification. However, their effects on the global flow patterns are distinct due to the nature of the individual terms — the Coriolis leads to latitudinal variations being stronger closer to the poles and weaker at the equator, while the drag depends only on the velocity magnitude. For this reason, we vary K by varying both rotation and drag individually, to study if contrast properties are actually significantly affected by them. Finally, we vary W by changing the instellation variation frequency. All permutations included, we have a suite of 200 simulations ($25 \times 2 \times 2 \times 2$), the results of which are presented below.

4.4. Typical Earth-like planets

In this section we present results for a typical tidally-locked Earth-analog. For all the simulations, the following values are kept constant: $g = 9.81 \text{ m/s}^{-2}$, $a = 6400 \text{ Km}$, $H = 2 \text{ Km}$ and $\tau_{\text{rad}} = 10 \text{ days}$. Based on the above, the wave timescale comes out to be 5.28 days. This gives a X value of 0.528. τ_{drag} is varied between 0.28 and 55.9 days keeping a slow rotation period of 351 days. Rotation period is varied from 1.76 to 351 days,

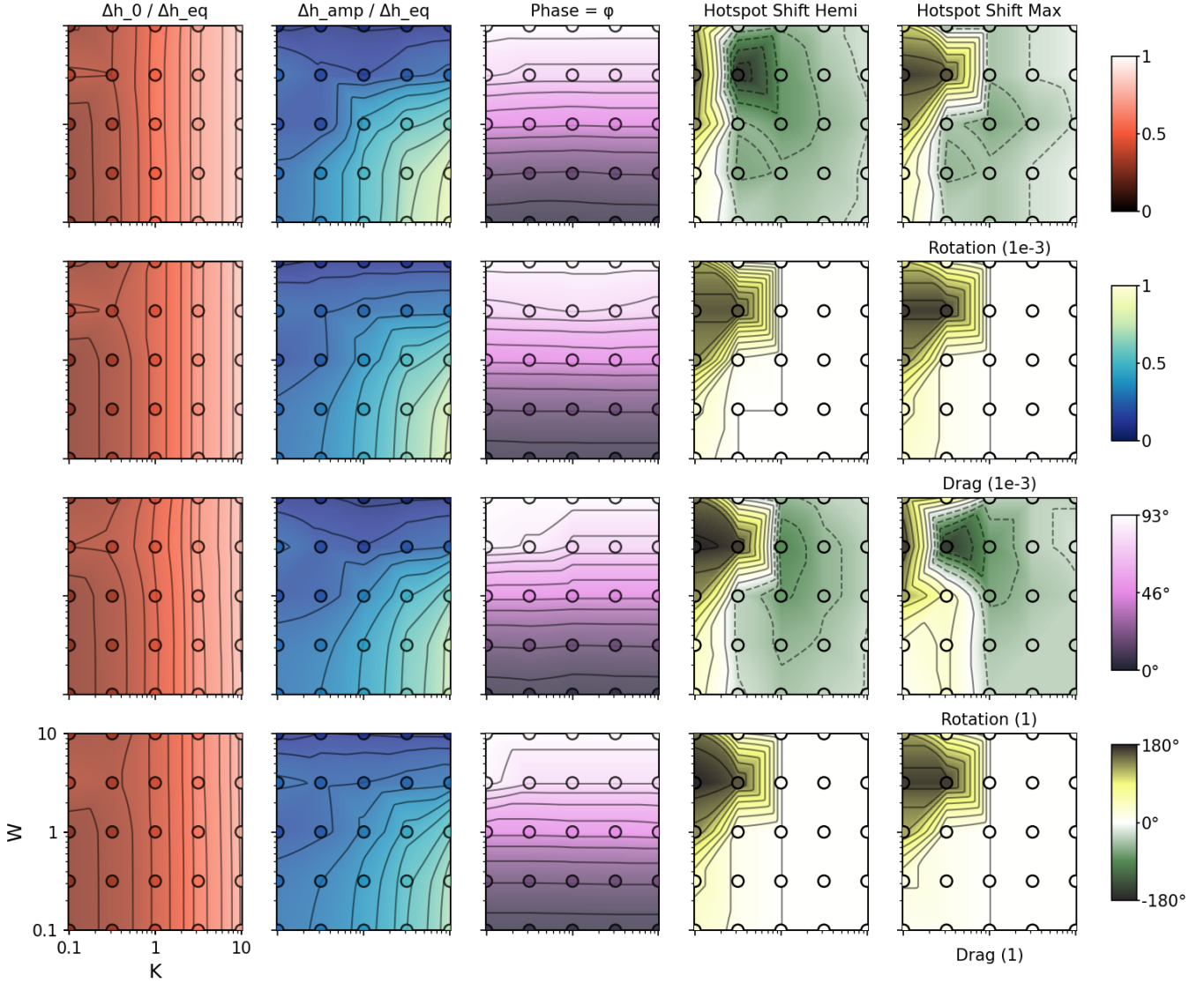


Figure 5. Suite of simulation results for an tidally-locked Earth-analog planet, showing the magnitude, amplitude, phase and maximum oscillating hotspot shift from the substellar point in the four columns, respectively. The colorschemes are identical to the ones used in theory plots in figure 4. The theoretical results capture the dominant features in the simulations including the second order resonance effects. Note that the magnitude plots feature a local maxima in frequency corresponding to the location of the resonance which is not seen in the theoretical results. Further, rotation and drag has overall similar effects for most quantities except for the hotspot shift (4th column).

keeping the drag timescale constant at 55.9 days. For these set of parameters, the systems are mostly in the linear regime, except close to $K = 0.1$ where the momentum equation loses linearity when $\Delta h_{eq} = 1 \times H$; see blue line in figure 2.

figure 5 shows the simulated planetary responses for the above set. The general trends match the order-one theoretical results. However, the region of resonance occurring at low K produces a local minima at $W = 2$, which is only a feature of the second-order theory. This effect is a little more pronounced in the bottom two rows where the forcing magnitude is three orders of

magnitude stronger than the top two rows. We also see the constant part of the response to have a manifestation of the resonance effect; **this is not captured by any of the linearized theories and possibly is due to higher order wave mean-flow interactions.**

An interesting phenomena is observed for some of the simulations when we track the location of the hotspot. The hotspot oscillates as it moves away from the substellar point towards the antistellar point and back proceeding from the *eastward* direction, leading to thermal variability not only in time, but also in space. When it happens, we find the hotspot to be oscillating at the

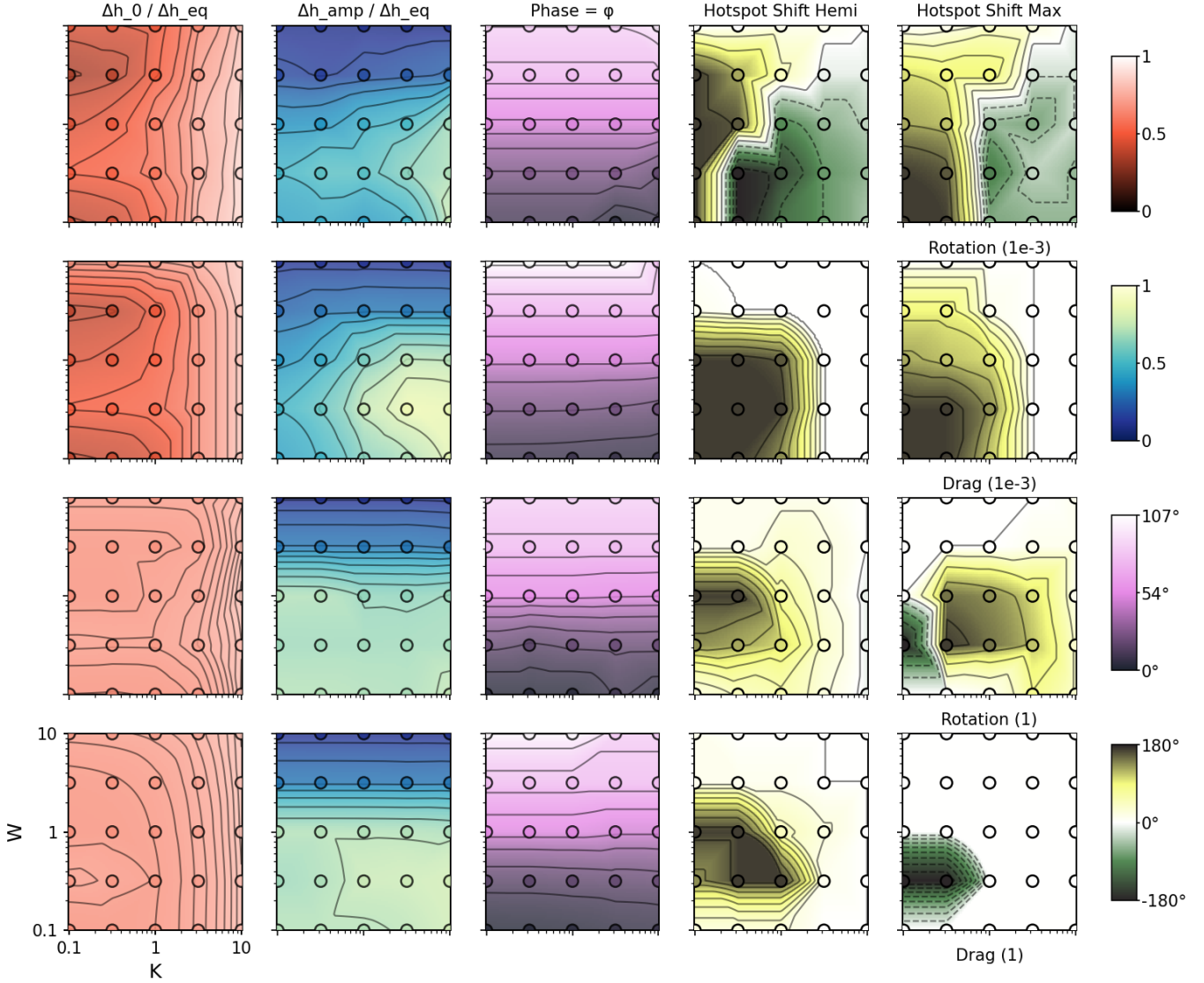


Figure 6. Suite of simulation results for an tidally-locked hot Jupiter planet, with the same quantities as in figure 5. Again, the theoretical results capture the dominant features in the simulations including the second order resonance effects. The magnitude plots still feature a local maxima in frequency corresponding to the location of the resonance which unseen in the theoretical results. Here, the differences between $\Delta h_{\text{eq}} = 10^{-3} \times H$ and $1 \times H$ are stark and can be observed in most quantities presented. Notably different is the loss of westward (negative) hotspot oscillations in the strongly nonlinear regime with $\Delta h_{\text{eq}} = 1H$ for cases with varying rotation (compare column 4, rows 1 and 3). This is likely because of the strengthening of the eastward equatorial jet caused by enhanced wave mean-flow interactions due to nonlinearities (cite papers here).

forcing frequency. The maximum amplitude of this spatial oscillation is plotted in the last column of figure 5. The oscillation is seen to correspond to the region of the $K - W$ parameter space where the second-order theory predicts the resonant maximization of temporal variability. This indicates that the phenomena are connected.

The primary difference between the cases with changing drag and rotation is also apparent in these plots. Both set systems show spatial amplitude maximization during the resonance. However, it is only the cases with changing rotation period that also feature *westward* os-

cillations. This behaviour is novel and is a key component of atmospheric variability for such planets, and will be investigated in detail in a later section.

4.5. Typical hot Jupiters

In this section we present results for a typical tidally-locked hot Jupiter with parameters from D. Perez-Becker & A. P. Showman (2013b). For all the simulations, the following values are kept constant: $g = 10 \text{ m/s}^{-2}$, $a = 82000 \text{ Km}$, $H = 400 \text{ Km}$ and $\tau_{\text{rad}} = 0.1 \text{ days}$. Based on the above, the wave timescale comes

out to be 0.3 days, which is much faster than the Earth-analog. Even the radiative timescale is significantly shorter. However, their ratio gives a higher value of $X = 3$, larger than the first set for which $X = 0.528$. τ_{drag} is varied between 0.22 and 45.04 days keeping a slow rotation period of 282 days. Rotation period is varied from 1.42 to 282 days, keeping the drag timescale constant at 45.04 days. Note that the rotation rate exploration goes way beyond the realm of hot Jupiters, but these cases are included for completeness. For these set of parameters, only the momentum equation is non-linear when Δh_{eq} is $1 \times H$, everything else is not. While effectively similar to the cases shown before, the degree of nonlinearity is much stronger; compare blue and red lines in figure 2. The continuity equation is always linear.

Figure 6 shows the planetary response to variable irradiation for the set of hot Jupiter planets described above. Comparing with the theoretical plots corresponding $X = 3$, we find a good match with both the first and second-order theories. The resonant frequency has now moved down to a smaller value of approximately $W = 0.25$. The top two rows for which Δh_{eq} is $10^{-3} \times H$, are in the linear regime, while the bottom two are not. The linear simulations match the theory better for obvious reasons — for the bottom rows, the $\Delta h_o / \Delta h_{\text{eq}}$ plots have more averaged out values over the $K - W$ parameter space, although the contour lines show that the fundamental structure is preserved. This is because, several of our assumptions breakdown (like what?) in the non-linear regime, and additional physics (like what?) comes in. This is also seen in the amplitude plots with values averaged out for low W values.

Why is phase never 180 degrees? The phase offset goes beyond 90° , but never to 180° as is determined theoretically. This is because

Finally, hotspot oscillations that appear in the resonant region are dominated by purely eastward movement in the nonlinear regime. The change can be seen comparing column 4 panels in rows 1 and 3. Westward oscillations still appear in at low forcing strengths as it is in the linear regime. This is different from what we have seen before and explained in detail in a latter section.

Effect of F on plots must be discussed.

Overall, our simulations across Earth-analog and hot Jupiter parameter regimes demonstrate good agreement between theoretical predictions and SWM results for the first and second-order resonance features. The key distinction between rotation and drag variations emerges in hotspot oscillation behavior: varying rotation produces both eastward and westward oscillations with a transition dependent on rotation rate, whereas varying drag

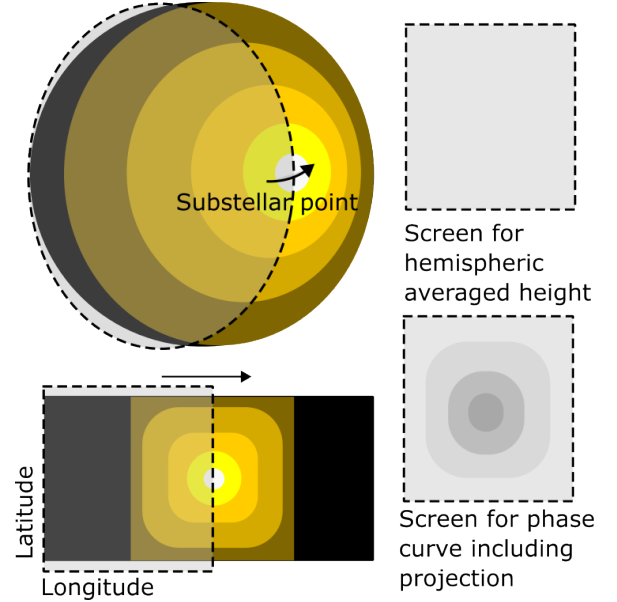


Figure 7. Typical equilibrium temperature distribution of a tidally-locked planet in two different representations, spherical and rectangular, showing the substellar point at the centre of the day side, and hemispheric screens for calculating the hemispheric averaged height and phase curves. The rectangular representation also shows the latitude, longitude and the night side. The arrows show the direction in which the screen moves to calculate the respective quantities. The screen for hemispheric averaged height is plain, while the second includes a projection to account for ray tracing effects needed for phase calculations.

yields only eastward oscillations. Linear regime simulations match theory more closely, while nonlinear regimes exhibit phase offsets that show preferentially eastward-dominated hotspot motion. These results validate the theoretical framework while highlighting important non-linear processes and rotation-dependent atmospheric dynamics that warrant further investigation.

5. APPLICATION TO REAL SYSTEMS

5.1. Variability

Show the contrast between westward and eastward oscillations. Compare constant instellation equivalent. Study those two cases, run them long, understand the dynamics. Comment on variability.

- Earth-like: drag cases have close to zero rotational component because of the low rotation rate of all runs. Rotational cases have significant rotational component. All in terms of KE.
- Hot-Jupiter like: does not reach statistical steady state in 10 rounds of instellation variations. Should run till max of (10 rounds or 40 days —

equivalent to the transient time). Above trends match.

- Figure 6 should be plotted for drag cases, and rotation cases separately, and also based on planet type — so four plots in all; compare hotspot offsets as well inside those plots as well
- It is the rotation rate that determines which way the oscillations occur. It transitions from eastward to westward with increasing rotation rate. When rotation stays constant and drag changes, the offset is always only eastward. In the non-linear regime, the all oscillations are eastward as well!
- I want to show how the hotspot oscillation behaviour changes with rotation rate. Overall, it switches from purely eastward to purely westward.

Or from purely eastward to zero. What is the mechanism?

- IDENTIFY and plot hovmollers for different unique hotspot behaviors. Earth: rotation 3, 4, and 13. Hot Jupiter: 2, 12, 22.
- Take FFT of phase curve to determine dominant frequencies for irradiation and response both. We see at least three frequencies currently, and the speculation is that something similar to beating happens.

5.1.1. Theory

5.2. Helmholtz decomposition

Software: Claude, Gemini, ChatGPT, NumPy, SciPy

APPENDIX

REFERENCES

- Banik, D. 2025, arXiv e-prints
- Banik, D., Menou, K., & Anders, E. H. 2025, *The Astrophysical Journal*, 991, 1
- Burns, K. J., Vasil, G. M., Oishi, J. S., Lecoanet, D., & Brown, B. P. 2020a, *Phys. Rev. Research*, 2, 023142
- Burns, K. J., Vasil, G. M., Oishi, J. S., Lecoanet, D., & Brown, B. P. 2020b, *Physical Review Research*, 2, 023068
- Dobrovolskis, A. R. 2009, *Icarus*, 204, 1
- Jones, K. D., Morris, B. M., Demory, B.-O., et al. 2022, *Astron. Astrophys.*, 666
- Kataria, T., Showman, A. P., Lewis, N. K., et al. 2013, *Astrophys. J.*, 767
- Komacek, T. D. 2025, *The Astrophysical Journal*, 983, 7
- Langton, J., & Laughlin, G. 2008, *Astrophys. J.*, 674, 1106
- Liu, B., & Showman, A. P. 2013, *The Astrophysical Journal*, 770, 42
- Ohno, K., & Zhang, X. 2019, *The Astrophysical Journal*, 874, 1
- Penn, J., & Vallis, G. K. 2017, *The Astrophysical Journal*, 842, 101
- Perez-Becker, D., & Showman, A. P. 2013a, *Astrophys. J.*, 776, 134
- Perez-Becker, D., & Showman, A. P. 2013b, *The Astrophysical Journal*, 776, 134
- Pierrehumbert, R., & Hammond, M. 2019, *Annu. Rev. Fluid Mech.*, 51, 275
- Showman, A. P., & Polvani, L. 2011, *The Astrophysical Journal*, 738
- Showman, A. P., & Polvani, L. M. 2010, *Geophysical Research Letters*, 37
- Showman, A. P., & Polvani, L. M. 2011, *Astrophys. J.*, 738, 71
- von Essen, C., Mallonn, M., Borre, C. C., et al. 2020, *Astron. Astrophys.*, 639

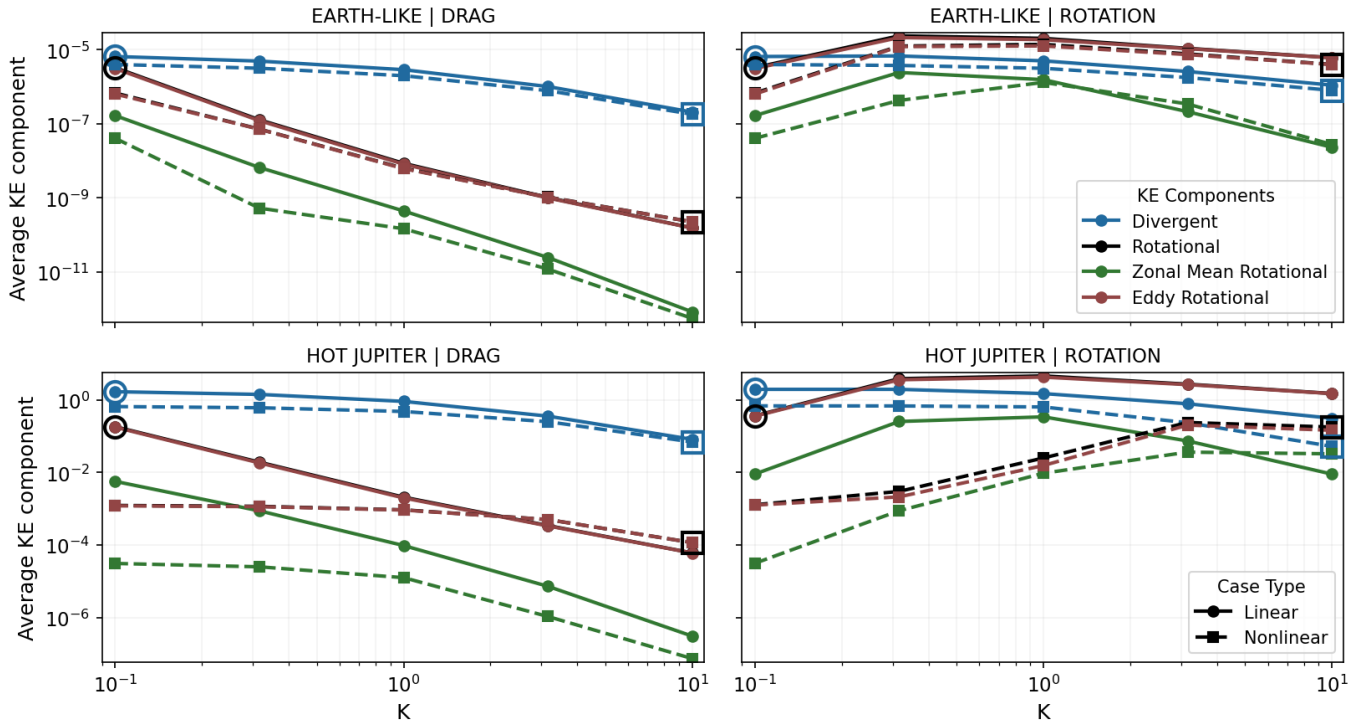


Figure 8. This figure does not have much to do with instellation variation. It would be the same if instellation did not change. It is more about the effect of rotation and drag on the flow patterns. Also, circle vs squares is not really linear vs nonlinear. Squares are closer to momentum nonlinearity. Earth-like: drag cases show decreasing trend for all KE components with increasing K because drag shows down winds. They also have significantly smaller rotational component because of the low rotation rate of all runs. Rotational cases show larger rotational components than drag cases. Hot-Jupiter like: the overall magnitudes of all KE are stronger, but similar trends as above are observed. **Question: should the average KE component determine the direction of hotspot offset?** As in, if the rotational KE is dominant, then the shift should be westward? This is seen in both rotation plots beyond a certain K value where the eddy rotational KE becomes greater than the divergent component. Eddy rotational and rotational are almost always overlapping, except in HJ-rotation. **Hot-Jupiter-rotation is the most nonlinear case:** even though the divergent contribution does not vary significantly between linear and nonlinear, the rotational does, most so for the HJ-rotation. What impact does this have on hotspot oscillations? Should we even keep this figure? Some of the cases have not reached a steady state in terms of KE.

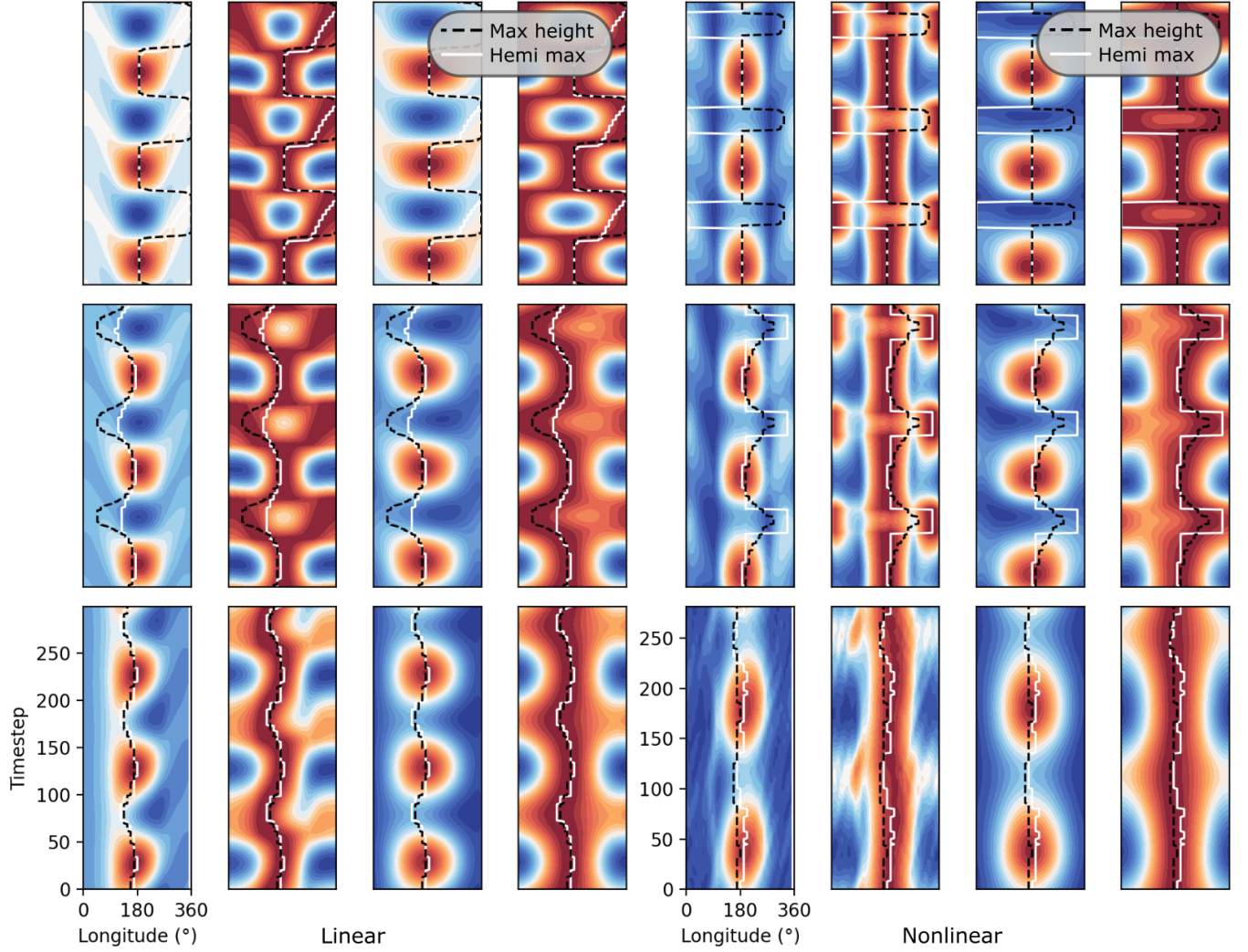


Figure 9. Hovmöller diagrams or timeseries of hotspot location relative to background zonal height profiles for six hot Jupiter planets showing hotspot oscillations switching from westward to eastward as the system transitions from linear to nonlinear regime. The left column corresponds to $\Delta h_{\text{eq}} = 0.001$ and the second column to $\Delta h_{\text{eq}} = 1$. Each row represents an increasing rotation period (282.9, 14.9 and 1.42 days), and columns show different diagnostic quantities: the meridional mean height, the meridional mean height normalized by the max height at each timestep to expose the instantaneous hotspot location, the hemispheric average height, and similarly normalized hemispheric average. The colorbars for background height fields are dropped to keep the figure clean focussing on the hotspot locations. White solid lines track the meridional mean-height hotspot, while black dashed lines track the hemispheric-maximum hotspot. The time window shown spans the last 3 forcing periods of the respective simulations.

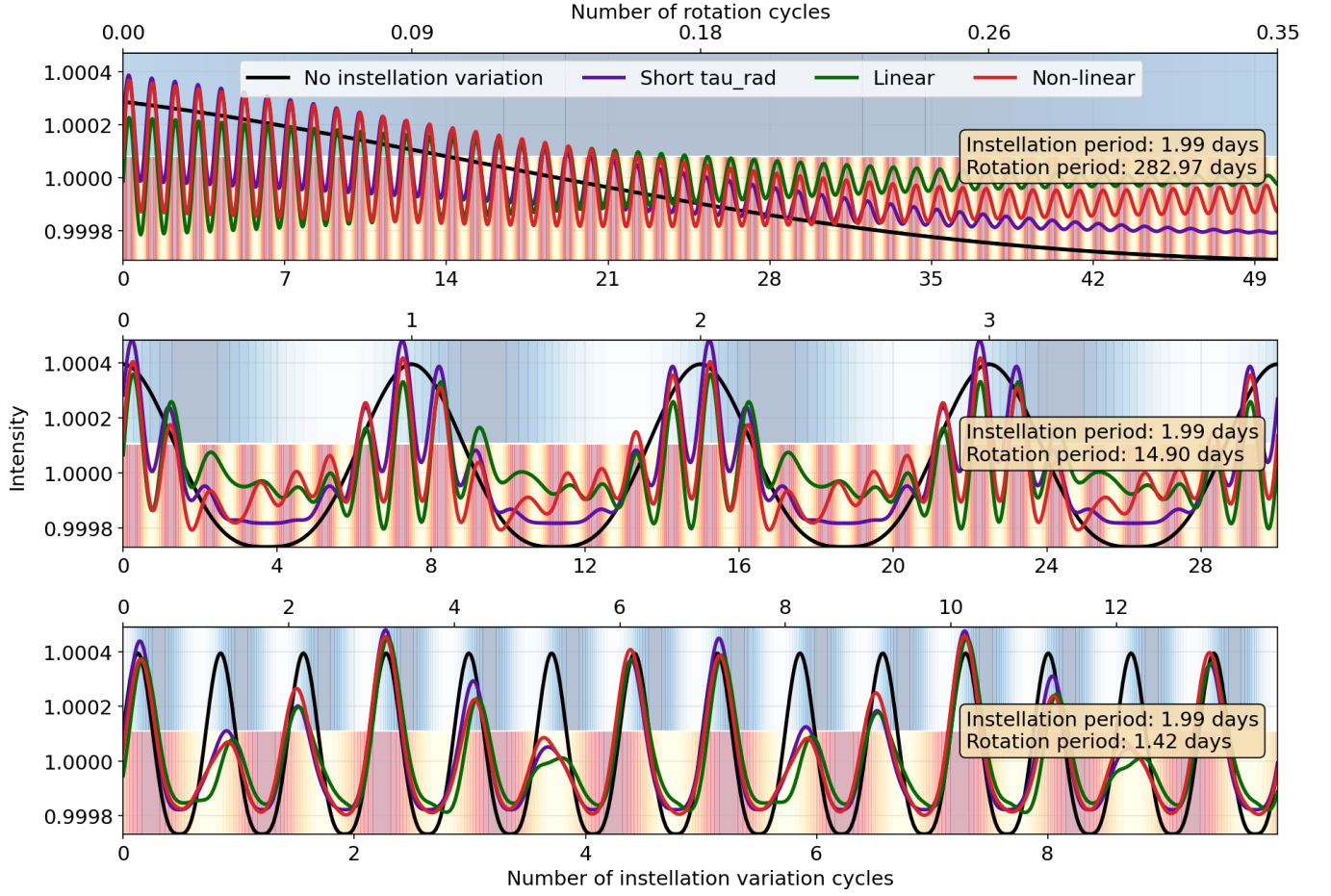


Figure 10. Phase curves for the same rotation periods as in Figure 9 with linear and nonlinear cases show significant differences. Each panel shows four normalized phase curves: case for no instellation variation, or simply the phase curve of the forcing profile (black); the same with instellation variation (purple); the linear case (green); and the nonlinear case (red). The background contours represent instellation variations (yellow-red, lower half) and rotation variations (blue and white, upper half) with corresponding cycle number on the top and bottom axes.